

LearnSphere EdTech

Technical Analysis Report

Data cleaning, SQL analysis, and Power BI dashboard build – methodology and technical findings

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Tools: Excel · MySQL · Power BI

1. Project Overview & Objectives

This report documents the technical methodology behind the LearnSphere analytics project: a simulated engagement for a South African online education company. The brief was to take an uncleaned operational export, validate and prepare it for analysis, build a relational database, answer a defined set of business questions in SQL, and deliver a self-service Power BI dashboard.

The pipeline followed three stages:

- Data preparation and quality control in Excel
- Schema design and analytical querying in MySQL
- Dashboard modelling and visualization in Power BI

2. Data Sources & Schema

The raw export (LearnSphere_original_dataset.xlsx) was normalized into six related tables, stored in the cleaned workbook and loaded into a MySQL database (learnsphere_analysis.sql).

Table	Contents	Keys
students	Demographics, signup source/device, city & province	student_id (PK)
courses	Course catalog: category, level, price (ZAR), instructor, active flag	course_id (PK), instructor_id (FK)
instructors	Instructor profile: specialty, experience, rating, city	instructor_id (PK)
enrollments	Enrollment status, completion %, payment amount/method, rating given	enrollment_id (PK), student_id (FK), course_id (FK)
reviews	Free-text review linked to a specific enrollment	review_id (PK), enrollment_id (FK)
lookup_table	Authoritative city → province mapping used for validation	city (lookup key)

Relationships: students 1→N enrollments; courses 1→N enrollments; instructors 1→N courses; enrollments 1→1 reviews (optional); students.city validated against lookup_table.city.

3. Data Cleaning Methodology

Cleaning was performed in Excel, with a dedicated Findings sheet documenting each data-quality issue and how it was resolved before the data was considered analysis-ready. Four categories of issues were addressed:

3.1 City-province mismatches

Student-submitted city and province values were cross-checked against the lookup_table reference. Records where the submitted province didn't match the expected province for that city were flagged (mismatch_province) and corrected to the expected value rather than dropped, preserving sample size for the marketing/location analysis.

3.2 Invalid ages

Age values outside a plausible working–adult range were identified and corrected or excluded, so downstream averages (e.g. average student age) aren't distorted by data–entry errors.

3.3 Malformed phone numbers

Phone number formatting was standardized; entries that couldn't be resolved to a valid South African number pattern were flagged in the quality log.

3.4 Inconsistent payment amounts

amount_paid_zar values were checked against course list price (price_zar). Values were retained as–is where they reflected legitimate discounting, which is analysed directly in Section 4.7. Genuine outliers (e.g. negative or zero payments on completed paid enrollments) were reviewed separately.

4. Database & SQL Analysis

The cleaned tables were loaded into a MySQL database (CREATE DATABASE LearnSphere;) and queried directly against the normalized schema – no flattening or pre–aggregation was done in Excel, so every metric below is reproducible from raw rows. The script begins with row–count sanity checks on all six tables before moving to analysis.

4.1 Signup & enrollment trend (3–year window)

A UNION ALL of yearly signup counts and yearly enrollment counts (each filtered to the last 3 years relative to the latest date in the respective table) is combined and grouped by year, giving a single trended view of top–of–funnel growth vs. conversion into paid enrollment.

Year	Signups	Enrollments
2024	217	1,194
2025	187	1,145
2026 (partial, to date)	96	554

2026 is a partial year at the time of extraction, so its lower totals reflect year–to–date activity rather than a decline. 2024 and 2025 show broadly stable signup volume (217 vs. 187) with enrollments consistently running at roughly 5–6x signups – i.e. the average student enrolls in multiple courses per signup year.

4.2 Revenue and dropout by category

Courses are joined to enrollments and grouped by category to rank total revenue (SUM(amount_paid_zar)) and separately to rank dropout rate (share of enrollments with status = 'Dropped'), surfacing categories that are commercially strong but operationally leaking students.

Category (by revenue)	Total Revenue
Mobile Development	R436,593.11
Data Science	R256,699.10
Digital Marketing	R76,171.83

Category (by dropout rate)	Dropout Rate
Project Management	13.68%
Mobile Development	13.56%
Business Analysis	13.33%

Mobile Development is notable for appearing in both lists – it is the top revenue category by a wide margin, but also has the second–highest dropout rate (13.56%). This makes it the single highest–priority category for retention work: a small reduction in dropout there converts directly into disproportionate revenue.

4.3 Completion rate – overall and by course

```
SELECT ROUND(AVG(status = 'Completed') * 100, 2) AS overall_course_completion_rate FROM enrollments; -- Result: 37.97
```

The same Boolean-average pattern is applied per course_title (joined to courses) with a HAVING completion_rate <= 30 filter to isolate underperforming courses for instructional or content review. Two courses fall below this threshold:

Course	Completion Rate
Content Writing Skills	26.45%
Google Ads Certification	25.44%

The overall completion rate (37.97%) reconciles exactly with the Power BI Course Performance page's Avg Completion Rate figure, confirming the dashboard measure and the SQL definition are aligned.

4.4 Acquisition channel quality

Students are joined to enrollments, filtered to status = 'Completed' AND amount_paid_zar > 0, and grouped by device_type and signup_source, ordered by count with LIMIT 1, to identify the single channel × device combination producing the most paying, completing students:

Device Type	Signup Source	Paying & Completed Students
Desktop	Google Search	63

This confirms, at the conversion level rather than just signup volume, that Desktop + Google Search is LearnSphere's strongest acquisition combination – consistent with Google Search and Desktop separately leading the Students & Marketing dashboard's raw signup and device metrics.

4.5 Geographic prioritization

Students are left-joined to enrollments and grouped by city and expected_province (the validated province from Section 3.1), ranked by total enrollments and then revenue, to shortlist the top locations for marketing spend:

City	Province	Revenue	Total Enrollments
Johannesburg	Gauteng	R626,164.44	701
Cape Town	Western Cape	R559,343.30	645
Durban	KwaZulu-Natal	R511,409.72	588
East London	Eastern Cape	R266,199.00	259
Bloemfontein	Free State	R263,626.95	273

Johannesburg (Gauteng), Cape Town (Western Cape), and Durban (KwaZulu-Natal) are the three clear leaders on both enrollment volume and revenue, together accounting for the large majority of tracked revenue across the top five locations. These three metros are the natural first targets for incremental marketing spending.

4.6 Rating vs. completion correlation

Average student_rating and completion rate are computed per course/instructor pair and compared. Finding: no strong correlation was observed – higher-rated courses did not consistently show higher completion rates. This implies rating and completion are driven by different factors and should be tracked as independent KPIs, not proxies for one another.

4.7 Discount impact on completion

Enrollments are split into 'Discounted' (amount_paid_zar < price_zar) vs. 'Full Price' using a CASE expression, then completion rate is compared between groups. Finding: the gap was only 1.16 percentage points, indicating discounting alone is not a meaningful lever for improving completion in this dataset.

4.8 Instructor leaderboard

Enrollments are joined through courses to instructors, grouped by instructor, and ranked by SUM(amount_paid_zar), with the instructor's profile rating (instructors.rating – a static attribute, not an average of student–given ratings) shown alongside:

Instructor	Revenue	Profile Rating
Monica Morton	R442,169.67	4.8
Connie Adams	R423,386.64	2.8
Phyllis Rodriguez	R279,045.90	0.0
Rhonda Mullins	R255,288.03	2.7
Kirsten Bowen	R229,359.02	4.2
Alisha Martin	R222,271.95	3.8
Tracy Garcia	R214,321.16	3.6
Benjamin Sanchez	R153,830.85	3.0
James Franklin	R145,678.05	3.0
April Bailey	R76,171.83	4.1

Monica Morton leads revenue (R442,169.67) and also carries a strong 4.8 profile rating. Phyllis Rodriguez (R279,045.90 revenue, 3rd place) has a profile rating of 0.0, flagging a likely missing/unset rating value worth data–quality follow–up rather than a genuine zero score. Note this profile rating is distinct from the average student–given rating used in the Power BI Instructors page and in Section 4.6 – the two are calculated from different fields and will not always match the same instructor.

5. Power BI Dashboard Architecture

The MySQL schema (or the cleaned workbook, for portability) feeds a four–page Power BI report (LearnSphere.pbit), built as a star–style model with enrollments as the fact table and students, courses, and instructors as dimensions, joined via student_id, course_id, and instructor_id respectively.

5.1 Report pages

- Overview – top–line KPIs (students, enrollments, revenue, completion rate, avg rating) with YoY deltas, monthly enrollment trend, revenue by category, enrollment status breakdown, and a province signup map
- Course Performance – active course count, top revenue/dropout courses, completion rate by category and by course, price–vs–completion scatter, and category enrollment mix
- Students & Marketing – top signup source and device, average student age, average revenue per paying student, signup source volumes, device usage split, and a student–distribution map by province
- Instructors – instructor count, top–rated and top–revenue instructor callouts, a top 5 instructor table, top 10 revenue chart, and a completion–vs–rating scatter

5.2 Modelling notes

- Cross–filtering: Year, Province, Device Type, Category, Payment Method, and Signup Source are implemented as report–level slicers driving all visuals on a page
- Completion Rate is a measure over enrollments (share of status = 'Completed'), consistent with the SQL definition in Section 4.3, so dashboard and database figures reconcile
- Geographic visuals use city/province fields validated in the Excel cleaning stage (Section 3.1) to avoid map pins landing in the wrong province

6. Key Metric Definitions

Metric	Definition	Headline value
Completion Rate	Completed enrollments ÷ total enrollments	37.97% overall (SQL and dashboard agree)
Total Revenue	SUM(amount_paid_zar) across enrollments	R3.11M (+18.3% YoY)

Metric	Definition	Headline value
Avg Revenue per Paying Student	Total revenue ÷ distinct paying students	R5.27K (98.5% of students have paid)
Dropout Rate	Dropped enrollments ÷ total enrollments (per category/course)	Highest category: Project Management (13.68%)
Discounted Enrollment	amount paid zar < course price zar	Completion delta vs. full price: 1.16pp

7. Findings Summary

- No strong correlation between average student rating and completion rate – quality perception and follow-through are independent behaviours in this dataset.
- Discounting has a negligible effect on completion (1.16pp difference), so price alone is unlikely to be the primary lever for improving completion.
- Revenue concentration is uneven across categories and instructors – Mobile Development alone generates R436,593.11, more than any other category, and Monica Morton leads the instructor leaderboard at R442,169.67, roughly double the 10th-place instructor's revenue.
- Mobile Development is simultaneously the top revenue category and the 2nd-highest dropout category (13.56%) – the clearest single opportunity to protect revenue through retention work.
- Desktop is the leading device (40.7% of signups) and Google Search as the leading acquisition channel (20.7%). This suggests current marketing and UX priorities should stay desktop-first while device diversification is monitored.

8. Limitations & Assumptions

- The dataset is a bounded export (not a live feed), so trend figures reflect the period captured at export time rather than real-time performance.
- expected_province (used for geographic analysis) is a derived/validated field, not the raw student-submitted value – see Section 3.1.
- Correlation findings (Section 4.6) are descriptive, not causal; no statistical significance testing was applied given the analytical scope of this project.
- Discount depth (how much below price) was not segmented – only a binary discounted/full-price split was analysed.

9. Tools & Tech Stack

Tool	Role in this project
Excel	Data validation, cleaning, and quality-log documentation (Findings sheet)
MySQL	Relational schema, business-question SQL (learnsphere_analysis.sql)
Power BI Desktop	Data modelling, DAX measures, and four-page interactive report

Appendix: Business Questions Answered

- How have student signups and enrollment volumes trended over the last 3 years?
- Which course categories generate the most revenue, and which has the highest dropout rate?
- What is the overall course completion rate, and which courses/instructors underperform?
- Which signup source and device type bring in the most paying, completing students?
- Which cities/provinces should LearnSphere target for marketing spend?
- Does average student rating correlate with completion rate?
- What % of enrollments used for a discount, and how does that affect completion?
- Who are the top 10 instructors by revenue generated and by average rating?